



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.					
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Institute: (0203)SYMBIOSIS CENTRE FOR MANAGEMENT AND HUMAN RESOURCE DEVELOPMENT,
PUNE
Programme: (020343) MASTER OF BUSINESS ADMINISTRATION (BUSINESS ANALYTICS)

Batch: 2024-26
Semester: I
Course: Business Statistics
Course Code: 0203430105

Date: 15/03/2025

Maximum Marks: 40

Day: Saturday

Time: 09:30 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- 1) All questions are compulsory
- 2) Calculator is allowed

Q.1) The life time of a certain type of battery has a mean life of 400 hours and a standard deviation of 50 hours. Assuming normality for the distribution of life time, find

- (i) The percentage of batteries which have life time of more than 350 hours
(3 Marks)
- (ii) The life time value above which the best 25% of the batteries will have their life.
(3 Marks)
- (iii) The proportion of the batteries that have life time between 300 hours and 500 hours.
(4 Marks)

(CO1)

Q.2) A restaurant chain wants to compare the average spending of customers at two of its locations, Location X and Location Y. The management is considering increasing prices at Location X, but first, they need to determine if there is already a significant difference in customer spending between the two locations.

They collect a sample of customer bills (in dollars) from a random selection of 8 customers at each location over the past week.

Customer Spending Data (\$):

Location X:

24, 26, 25, 28, 22, 27, 29, 26

Location Y:

21, 23, 20, 22, 24, 19, 25, 20

- (i) Test whether the average customer spending at Location X is significantly higher than at Location Y at a 5% significance level. (5 Marks)
- (ii) If the results show a significant difference in spending, what other factors should management consider before raising prices at Location X? (5 Marks)

(CO2)

Q.3) A retail company is testing the effectiveness of three different promotional strategies to increase sales. The marketing team wants to determine whether there is a significant difference in the average sales generated by each promotional strategy. The company runs a 4-week trial of each promotion at different stores and records the weekly sales (in \$1000s).

The weekly sales data (in \$1000s) is as follows:

Promotional Strategy 1:

- 24, 26, 22, 25

Promotional Strategy 2:

- 27, 29, 28, 26

Promotional Strategy 3:

- 22, 23, 21, 20

The marketing team wants to use a one-way ANOVA to determine if there is a significant difference in the average sales between the three strategies at a 5% significance level.

- (i) State the null and alternative hypotheses for the one-way ANOVA test. (3 Marks)
- (ii) Given the following **ANOVA table**, compute missing values, interpret the results and determine if there is a statistically significant difference between the promotional strategies:

(i) Test whether the average customer spending at Location X is significantly higher than at Location Y at a 5% significance level. (5 Marks)

(ii) If the results show a significant difference in spending, what other factors should management consider before raising prices at Location X? (5 Marks)

(CO2)

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The marketing team wants to use a one-way ANOVA to determine if there is a significant difference in the average sales between the three strategies at a 5% significance level.

(i) State the null and alternative hypotheses for the one-way ANOVA test.

(3 Marks)

(ii) Given the following ANOVA table, compute missing values, interpret the results and determine if there is a statistically significant difference between the promotional strategies:

Source of Variation	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	F-statistic	p-value
Between Groups	94.67	?	?	17.72	0.005
Within Groups	?	?	2.67		
Total	118.67	11			

(iii) Based on the ANOVA output, should the marketing team reject the null hypothesis? (4 Marks)
What does this mean in the context of promotional strategies?

(3 Marks)
(CO4)

Q 4) A human resources department in a large corporation is investigating the relationship between employee satisfaction and work environment type (Open Office, Cubicles, Remote Work). They conduct a survey of 300 employees and categorize their satisfaction levels as Satisfied, Neutral, or Dissatisfied based on their current work environment.

The results are summarized in the following 3x3 contingency table:

Work Environment	Satisfied	Neutral	Dissatisfied	Total
Open Office	70	50	30	150
Cubicles	40	30	30	100
Remote Work	60	40	50	150
Total	170	120	110	400

(i) Determine if there is a significant relationship between work environment and employee satisfaction at 5% significance level.

(5 Marks)

(ii) If the HR department concludes that work environment and employee satisfaction are related, what actions could HR department consider to improve employee satisfaction?

(5 Marks)
(CO3)